

A Technical Introduction to Differential privacy machine learning

Section 1

Attacks in practice Because differential privacy techniques are implemented on real computers, they are vulnerable to various attacks not possible to compensate for solely in the mathematics of the techniques themselves. In addition to standard defects of software artifacts that can be identified using testing or fuzzing, implementations of differentially private mechanisms may suffer from the following vulnerabilities:

Section 2

Publications Calibrating noise to sensitivity in private data analysis, Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. 2006. In Proceedings of the Third conference on Theory of Cryptography (TCC'06). Springer-Verlag, Berlin, Heidelberg, 265–284. Calibrating Noise to Sensitivity in Private Data Analysis (This is the original publication of Differential Privacy, and not the eponymous article by Dwork that was published the same year.) Differential Privacy: A Survey of Results by Cynthia Dwork, Microsoft Research, April 2008 (Presents what was discovered during the first two years of research on differential privacy.) Differential Privacy: A Primer for a Non-Technical Audience, Alexandra Wood, Micah Altman, Aaron Bembenek, Mark Bun, Marco Gaboardi, et al, Vanderbilt Journal of Entertainment & Technology Law/Vanderbilt Journal of Entertainment, Volume 21, Issue 1, Fall 2018. (A good introductory document, but definitely *not* for non-technical audiences!) Technology Factsheet: Differential Privacy by Raina Gandhi and Amritha Jayanti, Belfer Center for Science and International Affairs, Fall 2020 Differential Privacy and the 2020 US Census, MIT Case Studies in Social and Ethical Responsibilities of Computing, no. Winter 2022 (January). Differential Privacy and the 2020 US Census. Garfinkel, Simson (2025). Differential Privacy. MIT Press Essential Knowledge. MIT Press. doi:10.7551/mitpress/15354.001.0001. ISBN 9780262551656. Bowen, Claire McKay and Simson Garfinkel, The Philosophy of Differential Privacy, AMS Notices, November 2021.

Section 3

Definition Let ϵ be a positive real number and A $\{\displaystyle \{\mathcal{A}\}\}$ be a randomized algorithm that takes a dataset as input (representing the actions of the trusted party holding the data). Let $\text{im } A$ $\{\displaystyle \{\text{trm } \{\text{im}\}\} \{\mathcal{A}\}\}$ denote the image of A $\{\displaystyle \{\mathcal{A}\}\}$. The algorithm A $\{\displaystyle \{\mathcal{A}\}\}$ is said to provide (ϵ, δ) -differential privacy if, for all datasets D_1 $\{\displaystyle \{D_{\{1\}}\}$ and D_2 $\{\displaystyle \{D_{\{2\}}\}$ that differ on a single

element (i.e., the data of one person), and all subsets S $\{\displaystyle S\}$ of $\text{im } A$ $\{\displaystyle \{\text{trm } \{\text{im}\}\} \{\mathcal{A}\}\}$:

Section 4

21st century In 2003, Kobbi Nissim and Irit Dinur demonstrated that it is impossible to publish arbitrary queries on a private statistical database without revealing some amount of private information, and that the entire information content of the database can be revealed by publishing the results of a surprisingly small number of random queries—far fewer than was implied by previous work. The general phenomenon is known as the Fundamental Law of Information Recovery, and its key insight, namely that in the most general case, privacy cannot be protected without injecting some amount of noise, led to development of differential privacy. In 2006, Cynthia Dwork, Frank McSherry, Kobbi Nissim and Adam D. Smith published an article formalizing the amount of noise that needed to be added and proposing a generalized mechanism for doing so. This paper also created the first formal definition of differential privacy. Their work was a co-recipient of the 2016 TCC Test-of-Time Award and the 2017 Gödel Prize. Since then, subsequent research has shown that there are many ways to produce very accurate statistics from the database while still ensuring high levels of privacy.

Section 5

2008: U.S. Census Bureau, for showing commuting patterns. 2014: Google's RAPPOR, for telemetry such as learning statistics about unwanted software hijacking users' settings. 2015: Google, for sharing historical traffic statistics. 2016: Apple iOS 10, for use in Intelligent personal assistant technology. 2017: Microsoft, for telemetry in Windows. 2020: Social Science One and Facebook, a 55 trillion cell dataset for researchers to learn about elections and democracy. 2021: The US Census Bureau uses differential privacy to release redistricting data from the 2020 Census.

Section 6

Example According to this definition, differential privacy is a condition on the release mechanism (i.e., the trusted party releasing information about the dataset) and not on the dataset itself. Intuitively, this means that for any two datasets that are similar, a given differentially private algorithm will behave approximately the same on both datasets. The definition gives a strong guarantee that presence or absence of an individual will not affect the final output of the algorithm significantly. For example, assume we have a database of medical records D_1 $\{\displaystyle D_{\{1\}}\}$ where each record is a pair (Name, X), where X $\{\displaystyle X\}$ is a Boolean denoting whether a person has diabetes or not. For example:

Section 7

ϵ -differentially private mechanisms Since differential privacy is a probabilistic concept, any differentially private mechanism is necessarily randomized. Some of these, like the Laplace mechanism, described below, rely on adding controlled noise to the function that we want to compute. Others, like the exponential mechanism and posterior sampling sample from a problem-dependent family of distributions instead. An important definition with respect to ϵ -differentially private mechanisms is sensitivity. Let d be a positive integer, D be a collection of datasets, and $f : D \rightarrow \mathbb{R}^d$ be a function. One definition of the sensitivity of a function, denoted Δf , can be defined by: $\Delta f = \max_{D_1, D_2 \in D} \|f(D_1) - f(D_2)\|_1$, where the maximum is over all pairs of datasets D_1 and D_2 in D differing in at most one element and $\|\cdot\|_1$ denotes the L1 norm. In the example of the medical database below, if we consider f to be the function Q_i , then the sensitivity of the function is one, since changing any one of the entries in the database causes the output of the function to change by either zero or one. This can be generalized to other metric spaces (measures of distance), and must be to make certain differentially private algorithms work, including adding noise from the Gaussian distribution (which requires the L2 norm) instead of the Laplace distribution. There are techniques (which are described below) using which we can create a differentially private algorithm for functions, with parameters that vary depending on their sensitivity.